# ES32079 Financial Derivatives
# Problem Set 3

## Question 1

LIBOR has been replaced in sterling markets by SONIA.

a) Explain briefly why LIBOR was phased out.

b) Explain how SONIA differs from LIBOR in terms of:

- the maturity of the underlying borrowing;
- the data used to calculate the rate; and
- the amount of bank credit risk included in the rate.

c) Why might replacing LIBOR with SONIA change the value of an existing interest-rate derivative?

## Question 2

A bank quotes an annual interest rate of 10% with semiannual compounding.

a) What is the interest rate for each six-month period?

b) Calculate the effective annual rate.

c) Calculate the equivalent continuously compounded annual rate using

$$r = m \ln\left(1 + \frac{R_m}{m}\right).$$

d) Explain why the quoted rate, effective annual rate and continuously compounded rate have different numerical values even though they represent the same investment return.

# Question 3

A two-year bond has a face value of \$100 and pays a 6% annual coupon semiannually. Each coupon payment is therefore \$3.

The continuously compounded zero rates are:

| **Time** $t_i$ | **Cash flow** $C_i$ | **Zero rate** $R(t_i)$ |
|---|---|---|
| 0.5 | \$3 | 5.0% |
| 1.0 | \$3 | 5.8% |
| 1.5 | \$3 | 6.4% |
| 2.0 | \$103 | 6.8% |

a) Explain why the final cash flow is \$103.

b) Calculate the present value of each cash flow using

$$PV(C_i) = C_i e^{-R(t_i)t_i}.$$

c) Calculate the price of the bond.

d) Explain why a different zero rate is used for each cash flow.

# Question 4

The continuously compounded zero rates are 3% for one year and 4% for two years.

a) Calculate the one-year forward rate applying between Year 1 and Year 2 using

$$R_F(1,2) = \frac{R(2) \times 2 - R(1) \times 1}{2 - 1}.$$

b) Show that the following two investment strategies produce the same value at the end of Year 2:

(i) investing for two years at the two-year zero rate;

(ii) investing for one year at the one-year zero rate and then investing for another year at the forward rate.

c) Is the forward rate necessarily equal to the one-year market rate that will be observed in one year's time? Explain briefly.

# Question 5

A forward rate agreement covers a six-month interest period beginning in one year. The notional principal is \$1,000,000 and the fixed rate agreed in the FRA is 5%.

At the beginning of the interest period, the market reference rate is 6%. Assume that the year fraction is

$$\delta = 0.5.$$

Consider the party that receives fixed and pays floating.

a) Calculate the FRA cash flow at the end of the interest period using

$$L(R_K - R_M)\delta.$$

b) State whether this party makes or receives the payment.

c) If the FRA settles at the beginning of the interest period, calculate the amount actually paid using

$$\frac{L(R_K - R_M)\delta}{1 + R_M\delta}.$$

d) Explain why the settlement amount is smaller in absolute value when it is paid at the beginning of the interest period.

# Question 6

A bond has a price of \$100, a duration of 4.2 and a convexity of 22. The bond's yield increases by 10 basis points.

a) Express the change in yield, $\Delta y$, in decimal form.

b) Estimate the percentage change in the bond price using duration:

$$\frac{\Delta B}{B} \approx -D\Delta y.$$

c) Improve the estimate using duration and convexity:

$$\frac{\Delta B}{B} \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2.$$

d) Explain why a rise in yield causes the bond price to fall.

e) Briefly explain why matching the duration of assets and liabilities does not eliminate every type of risk.

# Question 7

You enter a long futures position on an index at an initial futures price of 5,000. The contract multiplier is GBP 10 per index point.

| Day | Futures price |
|---|---|
| 0 | 5,000 |
| 1 | 5,050 |
| 2 | 4,980 |

a) Calculate the gain or loss on Day 1.

b) Calculate the gain or loss on Day 2.

c) Calculate the cumulative profit or loss at the end of Day 2.

d) Explain how daily settlement affects the timing of cash flows compared with a forward contract.

# Question 8

A non-dividend-paying share is currently priced at GBP 100. The continuously compounded risk-free rate is 5% per annum and a forward contract matures in nine months.

a) Calculate the theoretical forward price using

$$F_0 = S_0 e^{rT}.$$

b) Suppose the market forward price is GBP 106. Compare this price with the theoretical forward price.

c) Does an arbitrage opportunity exist? If so, state whether the arbitrageur should buy or sell the forward contract.

d) Briefly describe the other transactions required to exploit the price difference and calculate the arbitrage profit at maturity.